

Back to likelihood function

$$p(d|h) \propto e^{-\frac{1}{2}(d-h|d-h)}$$

the [↑] probability of the data given the model

→ assumptions: noise is gaussian

$$(A|B) = \frac{1}{T} \sum \frac{A(f) B^*(f)}{S_n(f)}$$

PSD of noise (on avg. how much noise is in each bin)

$$SNR^2 = \rho^2 = (h|h) \quad \text{inner product of signal with itself}$$

Now, deep dive on how this formulation gets used on ground-based

$$\tilde{h}(f) \rightarrow \tilde{h}(f; \vec{\theta})$$

← vector of parameters

bhb is numbers

$$\vec{\theta} \rightarrow [m_1, m_2, \vec{s}_1, \vec{s}_2, \vec{L}, \vec{K}, t_{\text{merge}}]$$

3D location in space

orientation of the orbit
3 Euler angles as seen by you

now we rewrite the likelihood (log likelihood)

$$\log p(d|\vec{\theta}) = -\frac{1}{2}(d-h(\vec{\theta})|d-h(\vec{\theta})) + \log \text{normalization} \dots$$

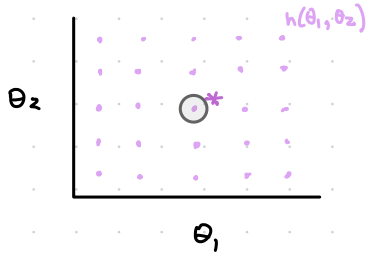
This $h(\vec{\theta})$ will maximize the log likelihood when $h(\vec{\theta}) = h^T$

↑ template

↑ The true signal

The "game" is called match filtering or template matching which changes the template to best match the signal.

Now we can outline a template design for this (grossly oversimp.)
condensed down to two params



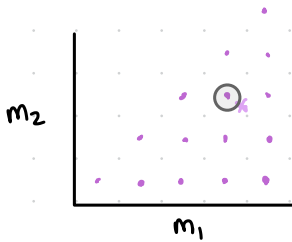
precalculated $h(\theta_1, \theta_2)$ on a grid
grav wave

- calculate likelihood at every value of the grid
- the highest value will be the closest to the detection

note on possible issues

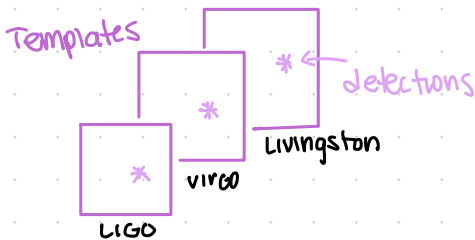
if grid is too sparse you may get multiple good options
grid too dense $\rightarrow$ too much computer power
bad model

ligo used the primary masses & ignore spin



- set convention so $m_1 > m_2$
so only half of the grid is needed

with multiple detectors

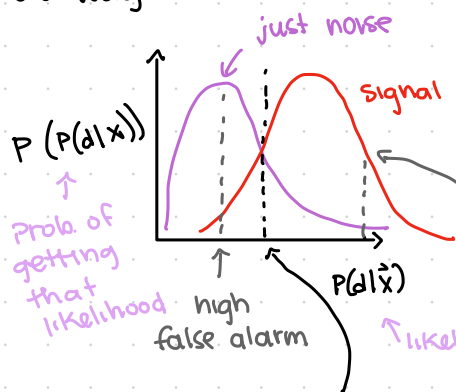


They look for

- 1) consistent masses (not identical)
- 2) time delay between detections is less than the light travel time between detectors

how do you decide if its a signal since the algorithm will always give a result?

one way



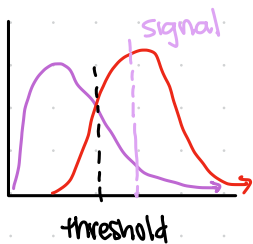
simulate a bunch of noise & feed it to the algorithm

feed it signal. it won't go as low as the noise.

confident it will be a signal but high false dismissals

choose a threshold of what is actually a signal
→ this one has some false positives & some false dismissals

maybe the signal population is shifted higher or lower
→ lots of assumptions required.



if we got this signal there is one more check of statistics of false positives.

LIGO can do this but not LISA

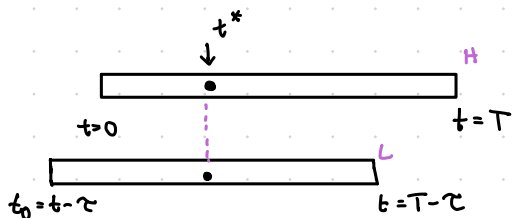
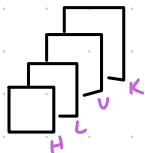
in real life

① we don't know the distributions of signals

② the noise statistics are a moving target

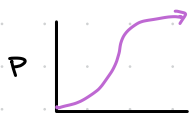
LIGO can empirically measure the background behavior of statistics in noise (darks & flats)

we leverage the consistency between detectors

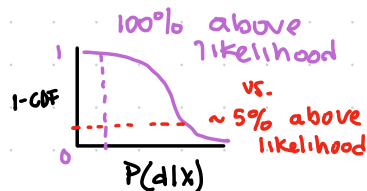


if $\tau \gg$ light travel we know its noise since the shift is larger than signal travel time

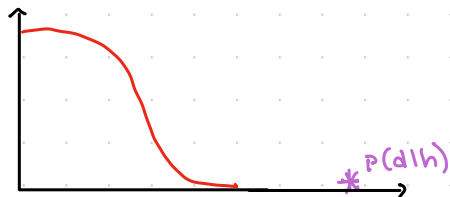
Try a *gazillion* shifts even knowing its just noise you can make a cumulative distribution function (CDF)



→ to a survival function 1-CDF



Now with lots of years of ^{noise} data (1 million) if you see a signal that's off the charts



the background has been estimated so we can say its detection rate is

$$F.A.R. < \frac{1}{1 \text{ million yrs}}$$

They can simulate years & years of noise data

one of the assumptions is that we never line up gravitational waves. LISA is signal dominated so time sliding won't work.

we also assumed only two parameters. If you have really long signals you need more dimensions.

This makes grid-based searches terrible for LISA

Back to LIGO

The search gives us a max likelihood value which corresponds to a point estimate $\{m_1, m_2\}$ which corr. to the template that hold those values. $\leadsto \{10M_\odot, 5M_\odot\}$

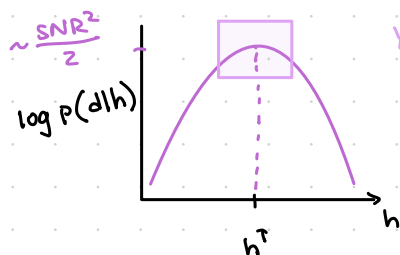
The paper would include

- False alarm rate
- point estimate
- max likelihood

but wait! $\{10M_\odot, 5M_\odot\}$ needs uncertainty

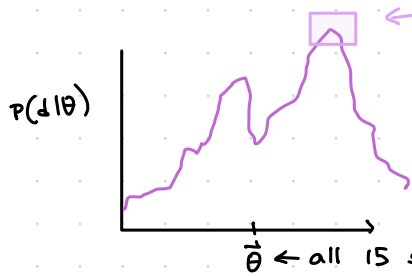
All the wrong ways to do it (the slow way)

* if its a modern paper do not tolerate this. if old give grace



you could say this region is pretty quadratic. get error over small region & call it good.

but we have 15 parameters.



← Probably pretty quadratic & assume errors are gaussian.

always true if you zoom in to a small enough patch

$\vec{\theta} \leftarrow$ all 15 squashed

This isn't great but it is pretty fast for high dim.

assume everything is gaussian

which makes the log likelihood quadratic @ max value

we approx the likelihood as

$$p(d|h) \sim e^{-\frac{1}{2}(d-h|d-h)} \Rightarrow p(d|\theta) \sim e^{-\frac{1}{2} \Gamma_{ij} \Delta \theta_i \Delta \theta_j}$$

↑
covariance matrix

which rewrites it in terms of parameters & a matrix rather than in terms of a template so we can fully characterize this knowing its point estimate & params

we can calculate Γ_{ij} with curvature

$$\Gamma_{ij} = \left(\frac{dh_i}{d\theta_i} \middle| \frac{dh_j}{d\theta_j} \right)$$



* recall gr. → curvature of the metric helps describe uncertainty

~~fisher~~ information matrix

he was a bad dude though so you can just say info. matrix.

This is the covariance matrix assuming gaussian parameters.

Toy example (two parameter model $\{x, y\}$)

↓ linear encoded waveform

$$\rightarrow h(x+y)$$

$$\rightarrow \left(\frac{dh}{dx} \middle| \frac{dh}{dy} \right) = 0$$

$$\Rightarrow \begin{pmatrix} \left(\frac{dh}{dx} \middle| \frac{dh}{dx} \right) & 0 \\ 0 & \left(\frac{dh}{dy} \middle| \frac{dh}{dy} \right) \end{pmatrix} = C^{-1}$$

inverse covariance for likelihood

$$\Rightarrow \begin{pmatrix} \left(\frac{dh}{dx} \right)^{-2} & 0 \\ 0 & \left(\frac{dh}{dy} \right)^{-2} \end{pmatrix} = C$$

some slight of hand

$$\begin{aligned} \rightarrow C_{xx} &\rightarrow \sigma_x^2 \\ C_{yy} &\rightarrow \sigma_y^2 \end{aligned}$$

- the diagonals are the variances
- off diagonals are how they are correlated which we said was 0 (linear)

So by def of Cov. matrix

$$\left(\frac{dh}{dx}\right)^2 = \frac{1}{\sigma_x^2}$$

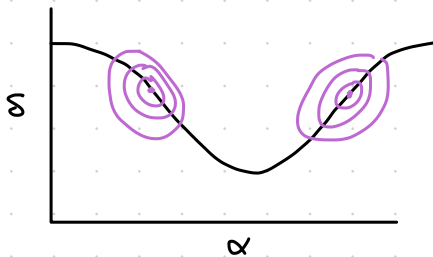
$$\left(\frac{dh}{dy}\right)^2 = \frac{1}{\sigma_y^2}$$

$$\Rightarrow \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{pmatrix}$$

which we can use to quantify the gaussian/quadr error.

one case this will always fail.

for ligo if the time delay is 1ms it makes a ring of equal time delay for all detectors



sky ring

you need more detectors to break the geometric symmetry

This method will only find one of these locations and assumes there is only one maximum value

Next time Bayesian Inference.